

Vena Pearl Boñolan, Ph.D.

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Room 313, Comp Sci and Eng. Lib 2.

Office Hours: Monday, Tuesday, Thursday, 2:30 – 6 pm or by appointment.

CS 294 Modeling

Prerequisite: a first course in differential equations, roughly CS 130 or ES 201;

A first course in numerical methods, roughly CS 131 or ES 204.

Credit: 3u

Applications of mathematical programming and modeling of engineered solutions involving, among others, multimedia environmental systems such as air pollution, groundwater flow and solute transport; stochastic models.

We will study the definition of modeling, simulation, and different methods of solution, including agent-based, FEM and cellular automata via ultradiscretization. We will also add stochastics in our models, and do sensitivity analyses of parameters.

Required downloads for this class: VenSim, NetLogo, FreeFem and Gama Platform. These are freely available on the net, and one of our objectives is to become familiar with their use.

There is no specific language requirement for this class, but it is expected that members should be able to modify or adapt existing code to meet requirements.

The elements of this course will be:

- a) Attendance. 10 %
- b) Problem presentations, roughly every other week, 30 %.
Ten points off from 100 for every week late.
- c) A midterm extended abstract of your work, to be submitted. 30 %.
This should incorporate at least two things learned in class. Ten points off 100 for every week late.
- d) A final research paper at least four pages. 30 %
This should incorporate at least three things learned in class. Ten points off 100 for every week late.

References:

“Integrated Environmental Modeling”, Rawaswami, A., Milford, J.B. and Small, M.J., 2005, Wiley and Sons. In library; very good resource for specific problems.

“Applied Mathematical Modelling of Engineering Problems”, Hritonenko, N.V., Yatsenko, Yuri P., 2003, Springer. In library.

“A First Course in Mathematical Modeling”, 4th. Ed., Giordano, F.R., Fox, W.P., Horton, S.B., Weir, M.D., 2009, Brooks/Cole. Two copies on reserve in libraries (Engg Lib 1 and 2).

“An Introduction to Scientific Computing, Twelve Computational Problems Solved with MatLab”, Danaila, I., Joly, P., Kaber, S.M., Postel, M., 2006, Springer. On reserve, Engg Lib 2.

“Mathematical Modelling with Case Studies”, Barnes, B. and Fulford, G.R., 2002, Taylor and Francis. Personal Book, Dr. Valdez.

Important Dates:

Midterms: Thursday, March 17

Last day of dropping Wednesday, 20 April

Last Day of Class: 19 May, Thursday

Finals: 21-28 May.